Deg4 partitions

Pablo Santos

October 17, 2026

Table 1: Table for $\lambda = (5,3,1,1)$, $\mu = (2,2,1,1)$:

$k = 6:$	$24V^{(4),\varnothing} \oplus 61V^{(3,1),\varnothing} \oplus 25V^{(2,2),\varnothing} \oplus 20V^{(2,1,1),\varnothing} \oplus V^{(1,1,1,1),\varnothing}$
$k = 5:$	$12V^{(5),(1)} \oplus 70V^{(4,1),(1)} \oplus 70V^{(3,2),(1)} \oplus 50V^{(3,1,1),(1)} \oplus 22V^{(2,2,1),(1)} \oplus 10V^{(2,1,1,1),(1)}$
$k = 4:$	$13V^{(5,1),(2)} \oplus 22V^{(5,1),(1,1)} \oplus 28V^{(4,2),(2)} \oplus 56V^{(4,2),(1,1)} \oplus 20V^{(4,1,1),(2)} \oplus 40V^{(4,1,1),(1,1)}$ $\oplus 13V^{(3,3),(2)} \oplus 22V^{(3,3),(1,1)} \oplus 20V^{(3,2,1),(2)} \oplus 40V^{(3,2,1),(1,1)} \oplus 7V^{(3,1,1,1),(2)}$ $\oplus 18V^{(3,1,1,1),(1,1)} \oplus 4V^{(2,2,1,1),(2)} \oplus 7V^{(2,2,1,1),(1,1)}$
$k = 3:$	$14V^{(5,2),(2,1)} \oplus 14V^{(5,2),(1,1,1)} \oplus 10V^{(5,1,1),(2,1)} \oplus 10V^{(5,1,1),(1,1,1)} \oplus 14V^{(4,3),(2,1)}$ $\oplus 14V^{(4,3),(1,1,1)} \oplus 24V^{(4,2,1),(2,1)} \oplus 24V^{(4,2,1),(1,1,1)} \oplus 10V^{(4,1,1,1),(2,1)}$ $\oplus 10V^{(4,1,1,1),(1,1,1)} \oplus 10V^{(3,3,1),(2,1)} \oplus 10V^{(3,3,1),(1,1,1)} \oplus 10V^{(3,2,1,1),(2,1)}$ $\oplus 10V^{(3,2,1,1),(1,1,1)}$
$k = 2:$	$3V^{(5,3),(2,2)} \oplus 4V^{(5,3),(2,1,1)} \oplus 3V^{(5,3),(1,1,1,1)} \oplus 4V^{(5,2,1),(2,2)} \oplus 8V^{(5,2,1),(2,1,1)}$ $\oplus 4V^{(5,2,1),(1,1,1,1)} \oplus V^{(5,1,1,1),(2,2)} \oplus 4V^{(5,1,1,1),(2,1,1)} \oplus V^{(5,1,1,1),(1,1,1,1)}$ $\oplus 4V^{(4,3,1),(2,2)} \oplus 8V^{(4,3,1),(2,1,1)} \oplus 4V^{(4,3,1),(1,1,1,1)} \oplus 4V^{(4,2,1,1),(2,2)}$ $\oplus 8V^{(4,2,1,1),(2,1,1)} \oplus 4V^{(4,2,1,1),(1,1,1,1)} \oplus V^{(3,3,1,1),(2,2)} \oplus 4V^{(3,3,1,1),(2,1,1)}$ $\oplus V^{(3,3,1,1),(1,1,1,1)}$
$k = 1:$	$2V^{(5,3,1),(2,2,1)} \oplus 2V^{(5,3,1),(2,1,1,1)} \oplus 2V^{(5,2,1,1),(2,2,1)} \oplus 2V^{(5,2,1,1),(2,1,1,1)}$ $\oplus 2V^{(4,3,1,1),(2,2,1)} \oplus 2V^{(4,3,1,1),(2,1,1,1)}$
$k = 0:$	$V^{(5,3,1,1),(2,2,1,1)}$

Table for $\lambda da = (4,4,4)$, $\mu = (12,)$, $k = 10$:
Table for $\lambda da = (5,3)$, $\mu = (4,2,1,1,1)$, $k = 10$:

$V^{(4,),(4,)}$
$2V^{(4,1),(5,)}$
$3V^{(4,2),(6,)} \oplus V^{(4,1,1),(6,)}$
$4V^{(4,3),(7,)} \oplus 2V^{(4,2,1),(7,)}$
$5V^{(4,4),(8,)} \oplus 3V^{(4,3,1),(8,)} \oplus V^{(4,2,2),(8,)}$
$4V^{(4,4,1),(9,)} \oplus 2V^{(4,3,2),(9,)}$
$3V^{(4,4,2),(10,)} \oplus V^{(4,3,3),(10,)}$
$2V^{(4,4,3),(11,)}$
$V^{(4,4,4),(12,)}$
$V^{(4,4,4),(12,)}$

$8V^{(2,),}$
$2V^{(3,),(4,)} \oplus 24V^{(3,),(3,1)} \oplus 14V^{(3,),(2,2)}$
$4V^{(4,),(4,1)} \oplus 4V^{(4,),(3,2)} \oplus 24V^{(4,),(3,1,1)} \oplus 16V^{(4,),(2,2,1)} \oplus 40V^{(4,),(2,1,1,1)} \oplus 12V^{(4,),(1,1,1,1,1)} \oplus 8V^{(3,1),(4,1)} \oplus$
$2V^{(5,),(4,1,1)} \oplus 2V^{(5,),(3,2,1)} \oplus 8V^{(5,),(3,1,1,1)} \oplus 6V^{(5,),(2,2,1,1)} \oplus 6V^{(5,),(2,1,1,1,1)} \oplus 2V^{(4,1),(4,2)} \oplus 10V^{(4,1),(4,1,}$
$V^{(5,1),(4,2,1)} \oplus 4V^{(5,1),(4,1,1,1)} \oplus 4V^{(5,1),(3,2,1,1)} \oplus 4V^{(5,1),(3,1,1,1,1)} \oplus 3V^{(5,1),(2,2,1,1,1)} \oplus 4V^{(4,}$
$2V^{(5,2),(4,2,1,1)} \oplus$